

Reference Exam

Surname(s):

Given name(s):

Student ID / Mat.-Nr.:

This is a practice exam with the same structure and scope as the actual exam. Use it to prepare.

	maximum	achieved
Question 1	15	
Question 2	35	
Question 3	35	
Question 4	15	
Total:	100	

Time allowed: 30 minutes Open book

DO NOT TURN OVER UNTIL INSTRUCTED!

Question 1 (15 points) Each of the following scenarios describes a real problem caused by a missing or misunderstood database concept. For each scenario: identify what went wrong, explain why it is a problem, and describe how a properly designed database would prevent it.

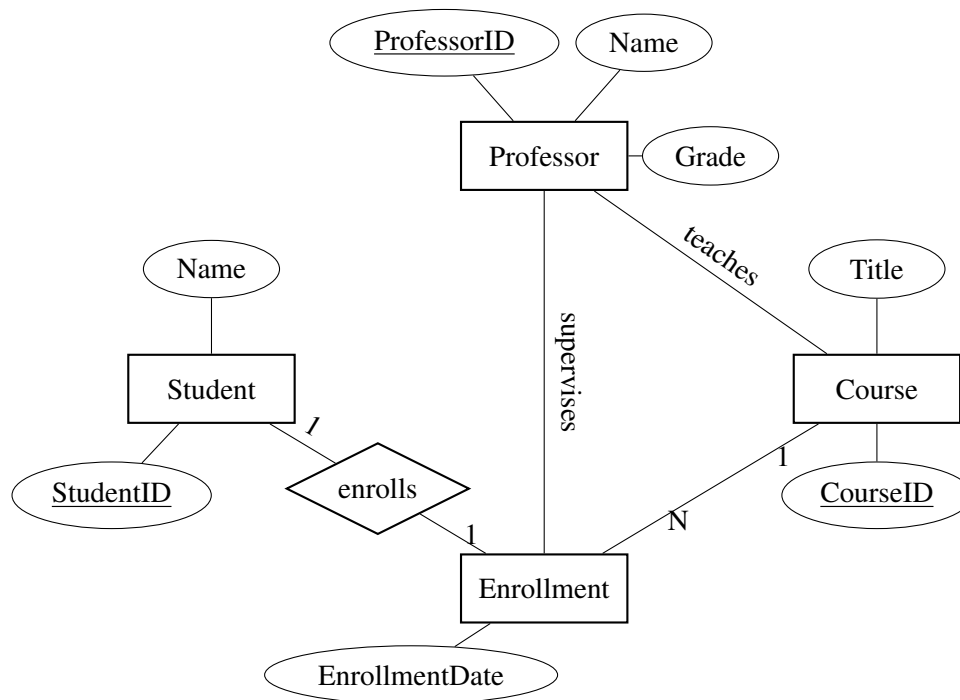
1. **(5 points)** A developer inserts a new order into the database:

```
INSERT INTO orders (order_id, customer_id, product_id, qty)
VALUES (1001, 999, 501, 2);
```

The database immediately rejects this statement with a constraint error. Customer ID 999 does not exist in the customers table.

- What type of constraint caused this rejection?
 - Is this rejection desirable behaviour? Explain what would happen to data quality over time if this constraint were not in place.
 - The developer suggests simply removing the constraint to “make the insert work.” What is wrong with this reasoning?
2. **(5 points)** A university database stores student enrolments in a table with columns `student_id`, `course_id`, and `grade`. The developer declares only `student_id` as the primary key. When the semester begins and students try to enrol in a second course, the system throws a duplicate key error.
- Why does this error occur?
 - What should the primary key be instead, and what is this type of primary key called?
 - Why is this the correct choice given the business logic of the table?
3. **(5 points)** A hospital system allows two nurses to work simultaneously. Nurse A starts updating a patient’s medication dosage from 10 mg to 20 mg but has not yet saved the change. Nurse B reads the same patient’s record at that moment and sees 20 mg. Nurse A then notices an error and cancels the update — the dosage reverts to 10 mg. Nurse B has already administered the 20 mg dose based on what she read.
- What is this concurrency problem called?
 - Which ACID property, and more specifically which isolation level, is designed to prevent it?
 - Why is this problem particularly dangerous in critical systems such as healthcare or banking?

Question 2 (35 points) The following ER diagram models a university course management system. It contains **four modeling errors**.



Business rules the diagram is supposed to capture:

- A student can enrol in many courses; each enrolment belongs to exactly one student.
- A course can have many enrolled students; each enrolment is for exactly one course.
- An enrolment records the date and the grade the student received.
- A professor teaches many courses; each course is taught by exactly one professor.

Tasks:

1. Identify **at least three** of the four errors in the diagram. For each error, state clearly what is wrong and why it violates correct ER modeling.
2. For the Student ↔ Enrollment relationship: what is the correct cardinality? Justify your answer using the business rules.
3. For the Professor ↔ Course relationship: what is the correct cardinality, and how should it be represented in an ER diagram?

Question 3 (35 points) A library stores all loan data in a single table:

LoanID	BookID	BookTitle	AuthorName	MemberID	MemberName	MemberEmail	LoanDate
1	B01	The Hobbit	J.R.R. Tolkien	M01	Alice	alice@lib.at	2026-01-10
2	B02	1984	G. Orwell	M01	Alice	alice@lib.at	2026-01-15
3	B01	The Hobbit	J.R.R. Tolkien	M02	Bob	bob@lib.at	2026-02-01

Business rules:

- A member can borrow many books over time.
- Each loan is for exactly one book.
- Each book has exactly one author.
- LoanID is the primary key.

Tasks:

1. **Normal form violations (15 points):** Which normal forms does this table violate? For each violation, name the normal form and explain *exactly why* it is violated — reference specific columns in your explanation.
2. **Anomalies (15 points):** Describe a concrete example of each of the following anomalies using this table:
 - **Insertion anomaly** — what data cannot be inserted, and why?
 - **Update anomaly** — what goes wrong when a value needs to change?
 - **Deletion anomaly** — what unintended information is lost when a row is deleted?
3. **Restructuring (5 points):** In plain words (no SQL required), describe how you would split this table to reach the Third Normal Form. Which tables would you create, and what columns would belong in each?

Question 4 (15 points) Part A — Transactions & ACID (8 points)

A hotel reservation system processes a booking in three steps:

1. Mark the selected room as “reserved” in the rooms table.
2. Create a booking record in the bookings table.
3. Deduct the deposit from the customer’s stored payment balance.

The database server crashes immediately after step 1, before steps 2 and 3 run.

Answer the following:

- Which ACID property guarantees that either all three steps are saved permanently or none of them are?
- After the server restarts, what state is the database in? Use the terms COMMIT and ROLLBACK in your explanation.
- Two customers simultaneously book the last available room. Both check availability and see “1 room available,” both proceed, and the room ends up double-booked. Which ACID letter is violated here, and what is the specific concurrency problem called?

Part B — Stored Procedures vs. Triggers (7 points)

A database administrator has two requirements:

- (a) Whenever a student’s grade is updated in the enrollments table, the course_statistics table must automatically be recalculated to reflect the new average grade for that course.
- (b) The registrar’s office needs a convenient way to retrieve all students enrolled in a given course — including their names and current grades — by providing a course ID.

For each requirement:

- State whether you would use a **Stored Procedure** or a **Trigger**, and explain why.
- In one sentence: what is the fundamental difference in *how* a stored procedure and a trigger are invoked?

